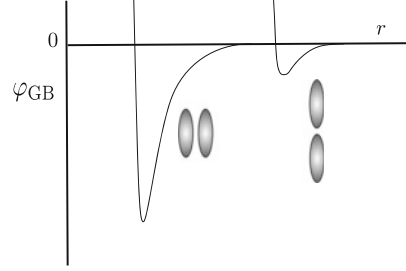


Fig. 1.12 The Gay-Berne potential depends strongly on the relative orientations of the rigid bodies, after a figure in [143]



in providing a precise parameterization [187]: using a general model for water (one not optimized for their specific experiment) the melting point of ice was underestimated by approximately 20 °K.

1.2 The N-Body Problem

In this book, we shall frequently use a compact, vectorial notation, where $\mathbf{q}$ and $\dot{\mathbf{q}}$ represent vectors of the positions and velocities, and $\mathbf{M}$ is a diagonal mass matrix, so the equations of motion (1.2) become

$$\mathbf{M} \frac{d^2}{dt^2} \mathbf{q} = \mathbf{F}(\mathbf{q}) = -\nabla U(\mathbf{q}). \quad (1.3)$$

If the N particles are confined to the real line, then $\mathbf{q} = (q_1, q_2, \dots, q_N)$ and the components of the force $\mathbf{F}$ are $F_i = -\partial U / \partial q_i$. For motion in $\mathbb{R}^3$, we write $\mathbf{q}_i = (q_{i,x}, q_{i,y}, q_{i,z})$ for the position vector of the i th particle, and the force acting on this particle is $\mathbf{F}_i = -\nabla_{\mathbf{q}_i} U = -\partial U / \partial \mathbf{q}_i$, that is, the vector with components $F_{i,x} = -\partial U / \partial q_{i,x}$, $F_{i,y} = -\partial U / \partial q_{i,y}$, $F_{i,z} = -\partial U / \partial q_{i,z}$. We may also write x_k, y_k, z_k for the x, y and z components of the position of particle k . We let N_c represent the total number of position coordinates of our system, thus for a system of N atoms in three space dimensions there are $N_c = 3N$ coordinates. In the typical case of 3-dimensional space, we view $\mathbf{q}$ as a $N_c = 3N$ -dimensional vector and write $\mathbf{M} = \text{diag}(m_1, m_1, m_1, m_2, m_2, m_2, \dots, m_N, m_N, m_N)$.

The number of local directions in which the configurational (position) state can be varied is called the number of *degrees of freedom* N_d . For the N -body system in $\mathbb{R}^3$ without additional constraints there are $N_d = N_c = 3N$ degrees of freedom. If r independent constraints are present the number of degrees of freedom is $N_d = N_c - r$.

The total energy of the N -body system is a function of positions and velocities,

$$E(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N, \dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2, \dots, \dot{\mathbf{q}}_N) = \sum_{j=1}^N \frac{m_j \|\dot{\mathbf{q}}_j\|^2}{2} + U(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N). \quad (1.4)$$

Along the solutions of (1.3), the energy is conserved, since its derivative vanishes:

$$\frac{d}{dt}E = \sum_{j=1}^N m_j \dot{\mathbf{q}}_j \cdot \ddot{\mathbf{q}}_j + \sum_{j=1}^N \frac{\partial U}{\partial \mathbf{q}_j} \cdot \dot{\mathbf{q}}_j = \sum_{j=1}^N \left(m_j \ddot{\mathbf{q}}_j + \frac{\partial U}{\partial \mathbf{q}_j} \right) \cdot \dot{\mathbf{q}}_j = 0.$$

As will become apparent in later chapters, it is sometimes necessary to introduce additional terms into the model to account for environmental conditions; these may destroy or alter the conservation of energy.

When we refer to the *value* of energy in a conservative system we typically write E instead of $\mathcal{E}$ (which represents the energy function).³ Thus the positions and velocities of system evolving under Newtonian dynamics satisfy the equation

$$E(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N, \dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2, \dots, \dot{\mathbf{q}}_N) = E.$$

In addition to the energy, there may be other constants of motion which arise from the structure of the equations of motion. For example, if the full set of particles of an N-body system is included in the model, then the instantaneous force acting on particle i due to the presence of particle j will be the exact negative of the force acting on particle j due to particle i ; this is a direct consequence of Newton's third law (every action has an equal and opposite reaction). This means that the sum of all the forces will vanish. Since

$$\frac{d\mathbf{p}_i}{dt} = \mathbf{F}_i,$$

we have

$$\sum_{i=1}^N \frac{d\mathbf{p}_i}{dt} = \sum_{i=1}^N \mathbf{F}_i = 0$$

and thus the three components of the *total momentum vector* $\mathbf{p}_{\text{tot}} := \sum_{i=1}^N \mathbf{p}_i$ will be conserved quantities. Additional first integrals sometimes arise in specific situations, depending on the form of the potential energy function.

For a few very special N-body systems, analytical solution is possible.

Example 1.3 (Harmonic Oscillator) Consider the system

$$\dot{x} = v, \quad \dot{v} = -\Omega^2 x.$$

³We use roman (non-italic) letters for parameters that describe the thermodynamic state, e.g. number of degrees of freedom, the volume, the energy, the temperature or the pressure.

Fig. 1.13 Table of units and their values for a model of argon (Ar)

Quantity	Unit	Typical value (Ar)
length	σ	3.4×10^{-10} m
mass	m	6.69×10^{-26} Kg
energy	ϵ	1.65×10^{-21} J
time	$\sigma\sqrt{m/\epsilon}$	2.17×10^{-12} s
velocity	$\sqrt{\epsilon/m}$	1.57×10^2 m/s
force	ϵ/σ	4.85×10^{-12} Newtons
temperature	ϵ/k_B	120K
pressure	ϵ/σ^3	4.20×10^7 Newtons/m ²

then the equations become

$$\frac{d^2 \mathbf{Q}_i}{d\tau^2} = - \sum_{\substack{j=1 \\ j \neq i}}^N \frac{\hat{\phi}'_{LJ}(R_{ij})}{R_{ij}} (\mathbf{Q}_i - \mathbf{Q}_j).$$

This means that a natural choice for the unit of time is

$$\alpha^{-1} = \sigma \sqrt{\frac{m}{\epsilon}} \approx 2.17 \times 10^{-12} \text{ s}.$$

By using the units given here, we may work with a simplified form of the Lennard-Jones system involving unit masses and a parameter-independent potential energy function.

In Fig. 1.13 we summarize the typical units relevant for studies of argon. The role of temperature and pressure will be discussed in Chaps. 6–8. $\square$

1.3 The Lagrangian and Generalized Coordinates

An N-body system is often formulated in terms of its *Lagrangian* $L = L(\mathbf{q}, \dot{\mathbf{q}})$: the difference between kinetic and potential energies. The somewhat mysterious Lagrangian function was introduced by Joseph Louis Lagrange. Its relevance becomes clear in formulating the *principle of least action*, a variational principle which gives an abstract formulation of the equations of motion. We defer introducing this derivation of the equations of motion until Chap. 2, where it is also used as a means of deriving a numerical method for molecular simulation.

For the system (1.3) with N atoms and $N_c = 3N$ configuration coordinates, the Lagrangian is

$$L \stackrel{\text{def}}{=} \frac{\dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}}{2} - U(\mathbf{q}).$$

The equations of motion may be expressed in terms of the Lagrangian as:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{q}}} = \frac{\partial L}{\partial \mathbf{q}}.$$

(Note that this must be interpreted in general as a set of $N_c = 3N$ equations, one for each atomic coordinate.) The real usefulness of the Lagrangian perspective comes when we introduce different types of coordinates to express the equations of motion; examples of this are given below.

When we introduce a smooth change of variables $\mathbf{q} = \boldsymbol{\Phi}(\mathbf{Q})$, where $\boldsymbol{\Phi} : \mathbb{R}^{N_c} \rightarrow \mathbb{R}^{N_c}$, this induces a corresponding transformation of the velocity vector by

$$\dot{\mathbf{q}} = \boldsymbol{\Phi}'(\mathbf{Q})\dot{\mathbf{Q}},$$

where $\boldsymbol{\Phi}'$ is the $N_c \times N_c$ Jacobian matrix of $\boldsymbol{\Phi}$. Then the Lagrangian of the system is transformed to

$$\tilde{L} = \frac{\dot{\mathbf{Q}}^T \boldsymbol{\Phi}'(\mathbf{Q})^T \mathbf{M} \boldsymbol{\Phi}'(\mathbf{Q}) \dot{\mathbf{Q}}}{2} - U(\boldsymbol{\Phi}(\mathbf{Q})),$$

It is not essential for the change of variables to be a map from $\mathbb{R}^{N_c}$ to $\mathbb{R}^{N_c}$. For example, in a system with constraints, the change of variables might represent a parameterization of the manifold defined by the constraints. Then the mapping $\boldsymbol{\Phi}$ might be from $\mathbb{R}^{N_d}$ to $\mathbb{R}^{N_c}$, where $N_c \geq N_d$. An abstract set of configuration variables is referred to as a set of *generalized coordinates*. The introduction of generalized coordinates in an N-body system typically leads to a Lagrangian which can be viewed as quadratic in the velocities but with a “configuration-dependent mass matrix.” We will assume that any such changes of variables are regular transformations in the sense that $\boldsymbol{\Phi}'$ is of full rank and the resulting generalized mass matrix is invertible.

Although generalized coordinates are often useful for understanding molecular systems, as we shall learn in later chapters, it is desirable to avoid the use of formulations with a configuration-dependent mass matrix since it complicates the design of numerical methods with good conservation properties. This is one reason that molecular simulation methods are typically described in a Cartesian coordinate framework.

1.4 Hamiltonian Formulation

In this section, we introduce the Hamiltonian description of the motion which is useful for discussing the underlying geometric structure associated to classical mechanics. We can understand the Hamiltonian formulation as a natural description of classical mechanics from several perspectives. First, it is based directly on the

most basic conservation law associated to a closed system (the conservation of energy) as it is formulated in terms of the energy function. Second, it is the starting point for the quantum mechanical treatment of a system of particles, since the Schrödinger equation is written in terms of the quantum analogue of the energy function. Third, we may view the Hamiltonian as the *Legendre transformation* of the Lagrangian viewed as a function of the velocities. This is a *conjugacy* relationship which ensures that the properties of the two systems are, under typical conditions, equivalent.

Abstractly, a Legendre transformation of a given convex function $g = g(\xi) : \mathbb{R}^m \rightarrow \mathbb{R}$ is a new function $\tilde{g} = \tilde{g}(\eta) : \mathbb{R}^m \rightarrow \mathbb{R}$ defined by

$$\tilde{g}(\eta) = \sup_{\xi} (\eta^T \xi - g(\xi)),$$

where the symbol $\sup_{\xi}$ means to find the supremum of the given expression with respect to variation of the vector ξ . In the case of the Lagrangian $L(\mathbf{q}, \mathbf{v}) = \mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2 - U(\mathbf{q})$ (which we have seen is the formulation of a mechanical system in generalized coordinates) we find

$$\sup_{\mathbf{v}} (\mathbf{p}^T \mathbf{v} - (\mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2 - U(\mathbf{q}))) = \sup_{\mathbf{v}} (\mathbf{p}^T \mathbf{v} - \mathbf{v}^T \mathbf{M}(\mathbf{q}) \mathbf{v} / 2) + U(\mathbf{q}).$$

If $\mathbf{M}(\mathbf{q})$ is invertible, the supremum is achieved precisely when

$$\mathbf{v} = \mathbf{M}^{-1}(\mathbf{q}) \mathbf{p},$$

which gives the standard definition of the momentum vector $\mathbf{p}$ in terms of velocities,

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{q}}} = \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}},$$

and, with $\mathbf{v} = \mathbf{M}^{-1}(\mathbf{q}) \mathbf{p}$, the Legendre transformation results in a new function

$$H(\mathbf{q}, \mathbf{p}) \stackrel{\text{def}}{=} \mathbf{p}^T \mathbf{M}(\mathbf{q})^{-1} \mathbf{p} / 2 + U(\mathbf{q}).$$

This is precisely the energy function, written in terms of positions and momenta. In this form it is termed the *Hamiltonian* and it provides an alternative description of the system. The equations of motion can be written

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}.$$

For constant $\mathbf{M}$ we obtain the dynamical equations,

$$\dot{\mathbf{q}} = \mathbf{M}^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = \mathbf{F} = -\partial U / \partial \mathbf{q}.$$